AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

Chemistry Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 6, 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

Empirical formula shows the simplest whole number ratio between atoms present in a compound.

Calculate the empirical formula of a compound that contains 22% carbon, 4.6 % hydrogen and 73.4 % bromine by mass. (Hint: Atomic mass of bromine =80 amu)

Space for Calculation

Q.2. (Total 5 Marks)

A sketch of periodic table series is given below.

Na	Mg	Al	Si	P	S	Cl	Ar

a. Predict the period number to which the given series of elements belongs. (1 Mark)

_____

_____

b. Which of the given elements has the smallest atomic radius? (1 Mark)

_____

_____

c. Which of the given elements shows the highest electronegativity? (1 Mark)

_____

_____

d. Which of the given elements will ionize easily? Give a reason for your answer. (2 Marks)

_____

_____

_____

_____

Q.3. (Total 6 Marks)

a. Magnesium chloride has a high melting and boiling point due to the presence of strong electrostatic force of attraction.

Explain why:

i. A magnesium ion has a charge of +2. (1 Mark)

_____

_____

ii The ions of magnesium chloride stay together . (1 Mark)

_____

_____

iii Magnesium chloride has no overall charge. (1 Mark)

_____

_____

iv. The formula of magnesium chloride is $MgCl_2$. (1 Mark)

_____

_____

b. When you come back from school and enter the kitchen, you smell the cooking aroma.

Name and define the process which makes it possible to smell the cooking aroma. (2 Marks)

_____

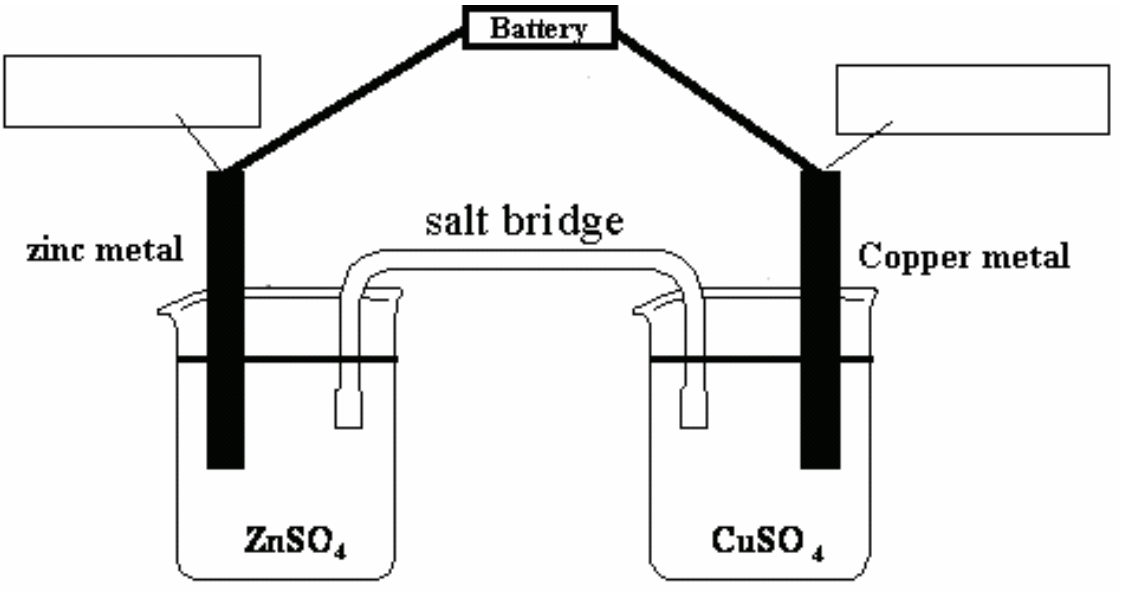
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Q.4. (Total 6 Marks)

a. A sketch of Daniel cell is given below.



i. Label the cathode and anode in the given diagram. (1 Mark)

ii. Show the flow of current with the help of arrows. (1 Mark)

iii. Identify at which electrode the oxidation occurs. (1 Mark)

_____

_____

iv. Give ONE use of the Daniel cell. (1 Mark)

_____

_____

b. Give the reason for the following cases. (2 Marks)

S.No	Cases	Reasons
i.	NaCl is soluble in HCl.	
ii.	Benzene is insoluble in water.	

Q.5. (Total 4 Marks)

a. Sodium is placed in group IA.

Support your answer by giving any TWO reasons. (2 Marks)

_____

_____

_____

_____

b. Write the complete balanced chemical equations for the following reactions.

i. Sodium reacts with cold water. (1 Mark)

_____

_____

ii. Sodium reacts with hydrochloric acid. (1 Mark)

_____

_____

Q.6. (Total 5 Marks)

EITHER

a. Write FIVE conclusions drawn by Rutherford after bombarding alpha particles from a gold foil.

OR

b. Classify mixtures and compounds from the given list. Based on the classification, give FOUR differences between mixtures and compounds.

Brass, Zinc sulphide, Soil, Water, Air

Q.7. (Total 5 Marks)

EITHER

a. Define covalent bond. Explain the THREE types of covalent bond with ONE example each.

OR

b. A bicycle pump contains $50\,cm^3$ of air at a pressure of $1 x 10^5$ pa .What would be the volume of the air if the pressure was increased to $2.1 x 10^5$ pa at constant temperature? Also state Boyle's law.

Q.8. (Total 5 Marks)

EITHER

a. A solution of sodium hydroxide is made by dissolving 22.5 grams of the salt in 500 cm^3 of the solution. Find the concentration of the solution in gm/dm^3 and mol/dm^3.

OR

b. Explain the electrolytic refining of copper with the help of construction, working and half cell reactions.

END OF PAPER

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